School of Computing

Graphic Era Hill University

Project Topic Proposal Form

1.Team Leader Details:

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- Section: A
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2.Project Details:

- Tentative Title of the Project: **“EmoSense AI:** Real-Time Facial Emotion Recognition System Using Deep Learning”
- Broad Area of Project: Artificial Intelligence / Deep Learning / Computer Vision

3.Problem Statement:

Human emotions play a critical role in communication and behavior analysis. However, traditional computer systems lack the ability to interpret and respond to human emotional states. There is a need for an intelligent system capable of detecting and classifying facial emotions in real-time using live webcam input. Such a system can be beneficial in applications like online education engagement monitoring, mental health assessment, customer experience analysis, and human-computer interaction.

This project aims to bridge this gap by developing a real-time facial emotion recognition system using deep learning techniques.

4.Objective and Significance:

The primary objective of this project is to design and implement a real-time web-based facial emotion recognition system using Convolutional Neural Networks (CNN).

The key objectives are:

- To preprocess and train a deep learning model on a facial emotion dataset.
- To classify seven human emotions: Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise.
- To integrate the trained model into a web-based application.
- To perform real-time emotion prediction using live webcam feed.
- To display emotion labels along with confidence percentage and emoji representation.

Significance of the project:

- Demonstrates practical implementation of deep learning and computer vision.
- Integrates AI with modern web technologies.
- Provides hands-on experience in model training, optimization, and deployment.
- Has real-world applications in AI-driven analytics, healthcare, surveillance, and education systems.

5.Expected Outcomes:

- A trained and optimized CNN-based emotion classification model.
- A real-time web application capable of:
  - Capturing live webcam feed.
  - Performing frame-by-frame emotion prediction.
  - Displaying emotion label with confidence score.
  - Showing corresponding emoji representation.
- A deployable frontend application using React and TensorFlow.js.
- Comprehensive documentation explaining architecture, methodology,

and results.

6.Tools, Technologies, and Methodologies Proposed:

Programming Languages:

- Python
- JavaScript

Frameworks & Libraries:

- TensorFlow / Keras
- TensorFlow.js
- React.js
- NumPy
- Matplotlib
- Google Colab (for GPU-based training)

Methodology:

1. Data Collection:
    - Using Kaggle Facial Emotion Recognition Dataset.
2. Data Preprocessing:
    - Grayscale conversion.
    - Image resizing to 48x48 pixels.
    - Normalization.
    - Data augmentation (rotation, flip, zoom).
3. Model Development:
    - Convolutional Neural Network (CNN).
    - Batch Normalization.
    - Dropout for regularization.
    - Softmax classifier for multi-class prediction.
4. Model Training:
    - GPU-based training using Google Colab.
    - Early stopping and model checkpointing.
5. Deployment:
    - Conversion of trained model to TensorFlow.js format.
    - Integration into React frontend.
    - Real-time inference in browser.

7.Project Type: Capstone Project

8.Mentor Details:

- Name of Mentor: Mohit Chauhan
- Mentor's Signature: ________________

9.Student Group Names and Signatures with Mobile numbers and email ids:

Team Leader:

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Team Member:

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